Multiple Choice Questions (MCQs) related to **Exception Handling in Java**,

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1. Which of the following is the superclass of all exceptions in Java?

A. `Error`
B. `Throwable`
C. `Exception`
D. `RuntimeException`
Answer: B. `Throwable`

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2. What will happen if an exception is not caught in a `try-catch` block?

A. The program will terminate abruptly
B. The program will continue execution
C. Java will ignore the exception
D. The code after the catch block will still execute
Answer: A. The program will terminate abruptly

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3. Which of the following exceptions is unchecked in Java?

A. `IOException`
B. `SQLException`
C. `NullPointerException`
D. `FileNotFoundException`
Answer: C. `NullPointerException`

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4. What is the purpose of the `finally` block in exception handling?

A. To handle only checked exceptions
B. To skip code execution after an exception
C. To execute code regardless of whether an exception occurs or not
D. To declare user-defined exceptions
Answer: C. To execute code regardless of whether an exception occurs or not

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5. Which of the following keywords is used to manually throw an exception in Java?

A. `try`
B. `catch`
C. `throw`

D. `throws`

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6. What type of exception is thrown when dividing an integer by zero?

A. `IOException`
B. `ArithmeticException`
C. `IllegalArgumentException`
D. `NumberFormatException`

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7. Which statement is true about checked exceptions in Java?

A. They can be ignored during compilation
B. They must be caught or declared in the method signature
C. They are subclasses of `RuntimeException`
D. They occur due to programming errors

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8. Which of the following statements correctly differentiates checked and unchecked exceptions in Java?

A. Checked exceptions are ignored at compile time, while unchecked exceptions must be handled
B. Checked exceptions must be either caught or declared in the method signature; unchecked exceptions do not require handling
C. Unchecked exceptions are subclasses of `Exception`, while checked exceptions are subclasses of `RuntimeException`
D. Unchecked exceptions must be handled using a try-catch block

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9. What will be the output of the following Java code?

Java
```java
public class Test {
    public static void main(String[] args) {
```

```
        try {
            int[] arr = {1, 2, 3};
            System.out.println(arr[5]);
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Exception caught");
        }
        System.out.println("Program continues...");
    }
}
```

A. Compilation error
B. Exception caught
Program continues...
C. Program terminates abruptly
D. ArrayIndexOutOfBoundsException is not caught

Unset

```
Exception caught
Program continues...
```

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10. What will be the output of the following Java code?

Java

```java
public class Example {
    public static void main(String[] args) {
        try {
            int result = divide(10, 0);
            System.out.println("Result: " + result);
        } catch (ArithmeticException e) {
            System.out.println("Cannot divide by zero");
        } finally {
            System.out.println("Finally block executed");
        }
```

```
    }

    public static int divide(int a, int b) {
        return a / b;
    }
}
```

A.

Unset
```
Result: 0
Finally block executed
```

B.

Unset
```
Cannot divide by zero
Finally block executed
```

C.

Unset
```
Cannot divide by zero
```

D. Compilation error due to division by zero

Unset
```
Cannot divide by zero
Finally block executed
```

https://www.youtube.com/watch?v=1XAfapkBQjk&t=89s